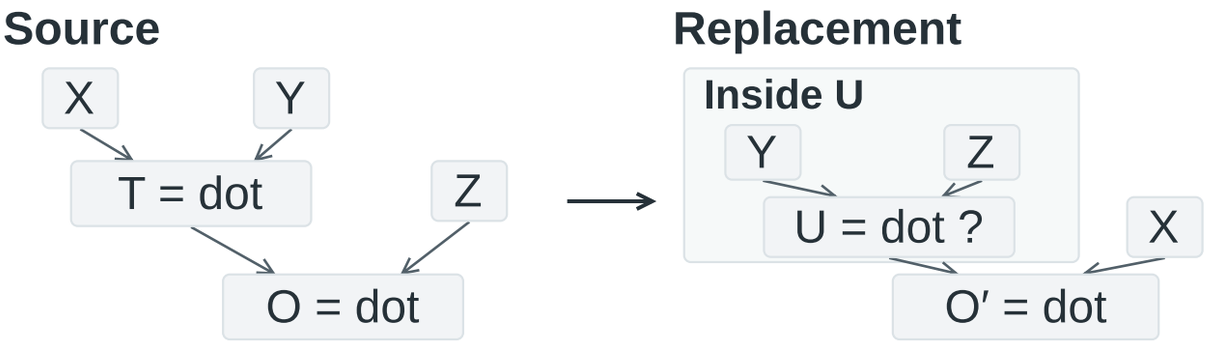



```
M_U = Axes_in(U) ∩
  (Axes_out(U) ∪ Axes(O))
```

Input Y
{ K | J_sel | J_rem }

Input Z
{ I_sel | J_sel | H }

Axes inside U: Y ∪ Z
{ K | I_sel | J_sel | J_rem | H }

Needed outside U?
X: { K | I_sel | I_rem }

Yes → **KEEP**
Used by X
{ K | I_sel }

Not needed here

Remaining candidates
{ J_sel | J_rem | H }

Required for output?
O: { I_rem | J_rem | H }

Yes → Required by O
{ J_rem | H }

∪

Not needed here

CONTRACT { J_sel }
Used only inside U

Inferred roles in U = dot(Y, Z)

Left free	Contract	Right free
from Y	shared by Y, Z	from Z
{ K | J_rem }	{ J_sel }	{ I_sel | H }

I_rem arrives via X. Axis order is not shown.
No batch axes; position lists are constructed afterward.